



SEEK WISDOM, ELEVATE YOUR INTELLECT AND SERVE HUMANITY !



School of public health

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Basics of Biostatistics

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Student's t -distribution

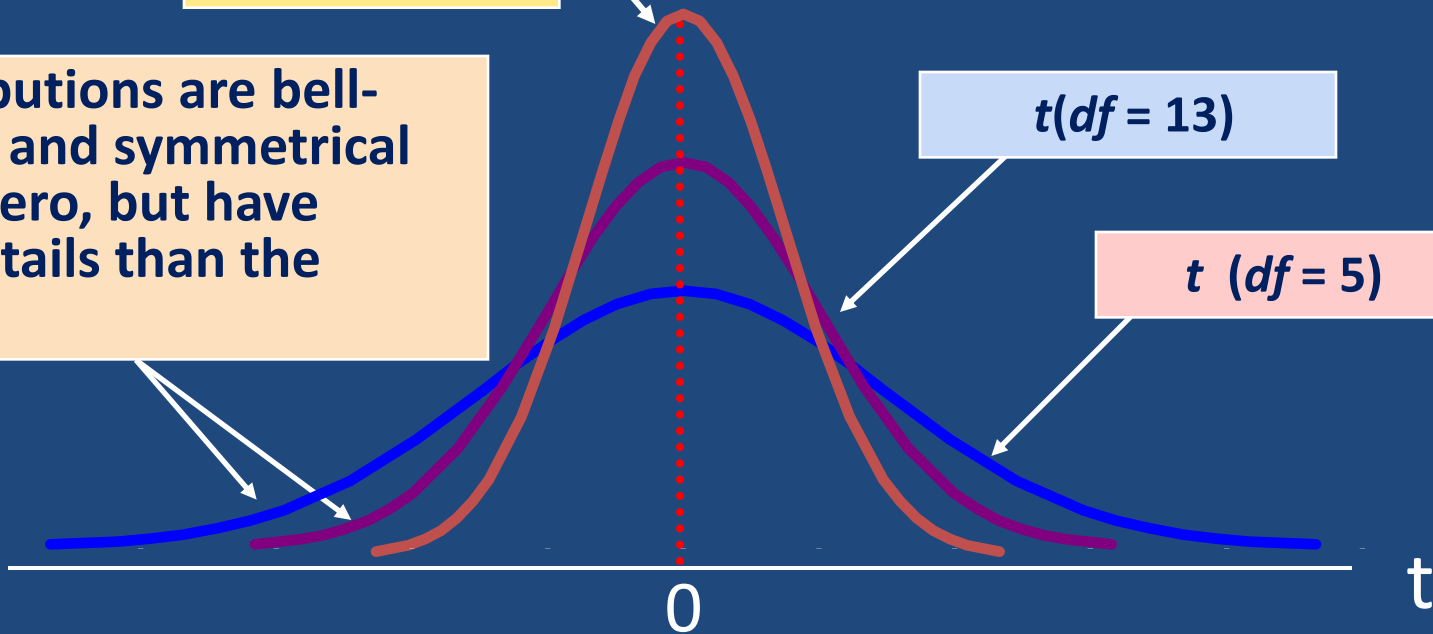
$t \longrightarrow Z$ as n increases

Standard
Normal
(t with $df = \infty$)

t -distributions are bell-shaped and symmetrical about zero, but have 'fatter' tails than the normal

$t(df = 13)$

$t(df = 5)$



Important Properties of the Student t Distribution

1. The Student t distribution is different for different sample sizes .
2. The Student t distribution has the same general bell shape as the normal distribution; its wider shape reflects the greater variability that is expected when s is used to estimate σ .
3. The Student t distribution has a mean of $t = 0$ (just as the standard normal distribution has a mean of $z = 0$).

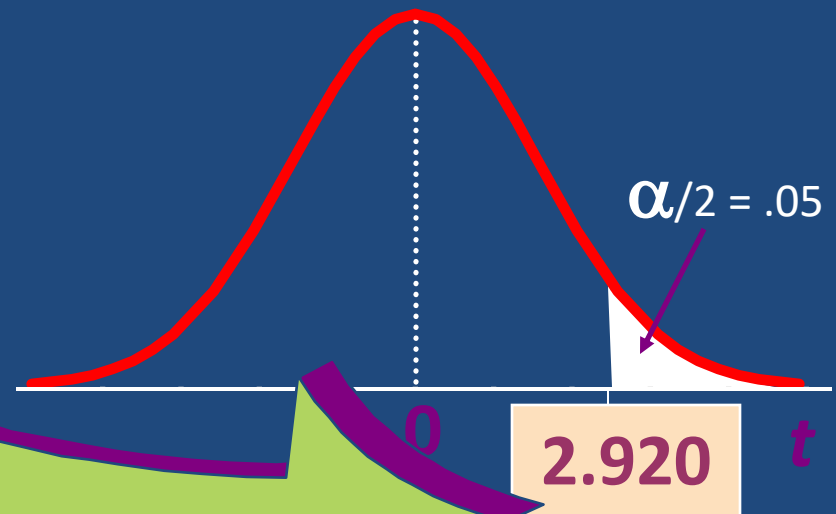
Important Properties of the Student t Distribution

4. The standard deviation of the Student t distribution varies with the sample size and is greater than 1 (unlike the standard normal distribution, which has $\sigma = 1$).
5. As the sample size n gets larger, the Student t distribution gets closer to the standard normal distribution.

Student's t Table

Right Tail Area			
df	.10	.05	.025
1	3.078	6.314	12.706
2	1.886	2.920	4.303
3	1.638	2.353	3.182

Let: $n = 3$
 $df = n - 1 = 2$
 $\alpha = .10$
 $\alpha/2 = .05$



The body of the table contains t values

t-distribution values With comparison to the Z value

<u>Confidence Level</u>	<u>$t_{\alpha/2}$ (10 d.f.)</u>	<u>$t_{\alpha/2}$ (20 d.f.)</u>	<u>$t_{\alpha/2}$ (30 d.f.)</u>	<u>$Z_{\alpha/2}$</u>
.80	1.372	1.325	1.310	1.282
.90	1.812	1.725	1.697	1.645
.95	2.228	2.086	2.042	1.960
.99	3.169	2.845	2.750	2.576

Note: $t \rightarrow Z$ as n increases

Estimating a single population mean

Margin of error(d)=reliability coefficient X standard error

$$d = Z_{\frac{\alpha}{2}} SE$$

$$= Z_{\frac{\alpha}{2}} \frac{\sigma}{\sqrt{n}}$$

$$\Rightarrow \sqrt{n} = Z_{\frac{\alpha}{2}} \frac{\sigma}{d}$$

$$\Rightarrow n = Z_{\frac{\alpha}{2}}^2 \frac{\sigma^2}{d^2}$$

11. To estimate the difference between two population means (Assuming that the two populations have common standard deviation σ)

$$n_1 = \left(1 + \frac{1}{r} \right) \frac{Z_{\frac{\alpha}{2}}^2 \sigma^2}{d^2}$$

- where n_1 is the size of sample 1 and r is the ratio of the size of sample 2 to sample 1. When the two samples are equal, the size of each sample is twice that given by one sample case.

II. Single population proportion

Let p denotes proportion of success, then

$$\begin{aligned}d &= Z_{\frac{\alpha}{2}} SE \\&= Z_{\frac{\alpha}{2}} \sqrt{\frac{P(1-P)}{n}} \\ \Rightarrow \sqrt{n} &= Z_{\frac{\alpha}{2}} \frac{\sqrt{P(1-P)}}{d} \\ \Rightarrow n &= Z_{\frac{\alpha}{2}}^2 \frac{P(1-P)}{d^2}\end{aligned}$$

Points to be considered

1. If sampling is from a finite population of size N , then

$$n = \frac{n_0}{\left(1 + \frac{n_0}{N}\right)}$$

where n_0 is the sample from an infinite population

2. The initial sample size approached in the study may need to be increased in accordance with the expected response rate, loss to follow up, lack of compliance, and any other predicted reasons for loss of subjects
3. Design effect for complex cluster sampling
Common values multiply n by 2, 3, ...5.

Using power of a study to determine sample size

IV. To test a hypothesis about the
difference between two population
means

(common standard deviation)

$$n_1 = \left(1 + \frac{1}{r}\right) \left(Z_{\frac{\alpha}{2}} + Z_{\beta}\right)^2 \frac{\sigma^2}{(\mu_1 - \mu_2)^2}$$

V. To test a hypothesis about the difference between two population proportions

■ Let
$$P = \frac{P_1 + rP_2}{1 + r}$$

$$n_1 = \frac{\left[Z_{\frac{\alpha}{2}} \sqrt{\left(1 + \frac{1}{r}\right) P(1-P)} + Z_{\beta} \sqrt{P_1(1-P_1) + \frac{P_2(1-P_2)}{r}} \right]^2}{(P_1 - P_2)^2}$$

♣ This equation is quite general: it applies to cross-sectional, cohort, and case-control study

Sample size based on relative precision

- The relative precision (ε) is defined as the absolute precision (d) divided by the value of the parameter.

- For a case-control study

$$n_1 = \left(\frac{Z_{\frac{\alpha}{2}}}{\ln(1-\varepsilon)} \right)^2 \left(\frac{1}{P_1(1-P_1)} + \frac{1}{P_2(1-P_2)} \right)$$

- where n_1 = the total number of cases
and

n_2 = the total number of
controls

- If the OR and one of the proportions are known, we can compute the unknown proportion by

$$P_1 = \frac{P_2}{P_2 + \frac{1 - P_2}{OR}}$$

- For comparing two proportion in case-control study

$$n_1 = \frac{\left[Z_{\alpha/2} \sqrt{\left(1 + \frac{1}{r}\right) \frac{1}{P(1-P)}} + Z_{\beta} \sqrt{\frac{1}{P_1(1-P_1)} + \frac{1}{r P_2(1-P_2)}} \right]^2}{(\ln OR)^2}$$

Examples

1. Suppose you wish to estimate the prevalence of acute respiratory tract infection, with a precision of 5%, in a target population comprising children aged 2-4 years in a particular region of a developing country. Since an estimate of P is not available until the survey has been carried out, the sample size calculation formula does not tell you what sample size is needed. However the formula may still be used to get a range of sample sizes corresponding to various assumptions for the values of P .

Solution

$$P = 0.1 \rightarrow n = (1.96)^2(0.1)(0.9)/(0.05)^2 = 138$$

$$P = 0.2 \rightarrow n = (1.96)^2(0.2)(0.8)/(0.05)^2 = 246$$

$$P = 0.3 \rightarrow n = (1.96)^2(0.3)(0.7)/(0.05)^2 = 323$$

$$P = 0.5 \rightarrow n = (1.96)^2(0.5)(0.5)/(0.05)^2 = 384$$


- ♣ For a fixed absolute precision d , the required sample size increases as P increases from 0 to 0.5, and then decreases in the same way as the prevalence approaches 1. If you had absolutely no idea about the prevalence (if you wished to estimate several different prevalences from the same survey), the safest approach to take is to assume that $P=0.5$.

2. A case-control study to compare the efficacy of a vaccine for the prevention of child-hood tuberculosis with a placebo. Assume that 30% of the controls are not vaccinated. If the numbers of cases and controls are equal what sample size is needed to detect, with 80% power and 5% type I error, an odds ratio of at least 2 in the target population?

Solution

- The probability of exposure (that is no vaccine given) for a control is $P_2 = 0.3$.

$$P_1 = 0.3 / (0.3 + 0.7/2) = 0.462$$

$$P = (0.3 + 0.462) / 2 = 0.381$$

$$n_1 = \frac{(1.96\sqrt{2 \times 0.381 \times 0.619} + 0.84\sqrt{0.462 \times 0.538 + 0.3 \times 0.7})^2}{(0.462 - 0.3)^2}$$
$$= 139.9$$

- Conclude that 140 cases and 140 controls are required for the study.

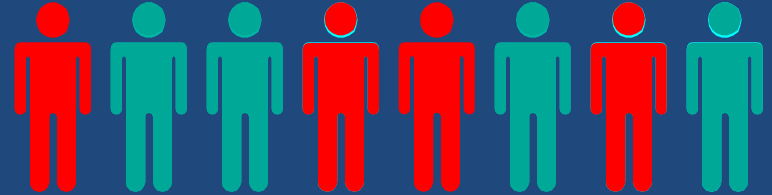
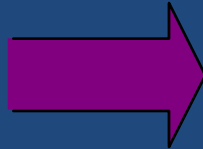
Hypothesis testing

- The purpose of hypothesis testing is to determine whether enough statistical evidence exists to enable us to conclude that a belief or hypothesis about a parameter is reasonable
- Examples
 - Is a new drug effective in curing a certain disease? A sample of patient is randomly selected. Half of them are given the new drug where half are given the standard drug . Then, the improvement in the patients conditions is measured and compared

Hypothesis Testing Process

Assume the
population
mean age is 50.

$$(H_0 : \mu = 50)$$



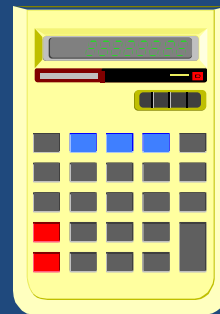
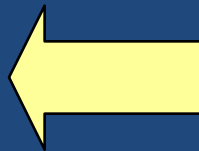
Identify the Population



Is $\bar{X} = 20$ likely if $\mu = 50$?

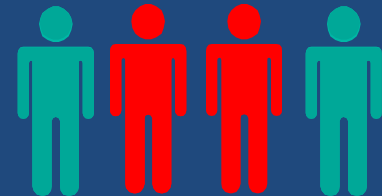
No, not likely!

REJECT H_0



$$(\bar{X} = 20)$$

Take a Sample



Steps in hypothesis testing

1) State the statistical hypotheses

- There are two hypotheses:
 - Null hypotheses
 - Alternative hypotheses

State the null hypotheses

- Null hypothesis – called the hypothesis of no difference or no association or no effect
- States that “there’s no difference” between the hypothesized value and the population parameter value
- Is always about a **population parameter**, not about a sample

Null Hypothesis: H_0

- The null hypothesis (denoted by H_0) is a statement that the value of a population parameter (such as proportion, mean, or standard deviation) is equal to some claimed value.
- Always contains the “=” “ $\leq$ ” or “ $\geq$ ” sign
- We test the null hypothesis directly.
- Either reject H_0 or fail to reject H_0 .

State the alternative hypotheses

- Alternate to null hypothesis
- Says'' there's a difference between the hypothesized value and the population parameter value
- It is what we are trying to prove, i.e. the reason for the research question.

Alternative Hypothesis: H_1 or H_A

- The **alternative hypothesis** (denoted by H_1 or H_A) is the statement that the parameter has a value that somehow differs from the null hypothesis.
- The symbolic form of the alternative hypothesis must use one of these symbols: $\neq$, $<$ or $>$.
- May or may not be accepted

Hypothesis

Example: Consider population mean

$$H_0: \mu = \mu_0$$
$$H_A: \mu \neq \mu_0$$

Two- tailed

$$H_0: \mu \leq \mu_0$$
$$H_A: \mu > \mu_0$$

One - tailed

$$H_0: \mu \geq \mu_0$$
$$H_A: \mu < \mu_0$$

One- tailed

2) Select the level of significance and determine the critical value

Level of Significance (α) : Defines rejection region of the sampling distribution

- Called rejection region of sampling distribution
- Is designated by α , (level of significance)
 - Typical values are .01, .05, or .10
- Is selected by the researcher at the beginning
- Provides the critical value(s) of the test

Determine the critical value

- Find the critical value in the distribution table for each test statistics(tabulated value) using the α - value and the degrees of freedom(where applicable)
- The critical value separates the **acceptance zone** from the **rejection zone**
- **One tailed test** – area of rejection is in either the lower or upper tail of the distribution
- **Two- tailed** – 2 areas of rejection, one in each tail of the distribution

Rejection Regions (Two -tail test)

Sampling Distribution

Level of Confidence

Rejection
Region

Rejection
Region

$1/2 \alpha$

$1 - \alpha$

$1/2 \alpha$

Nonrejection
Region

Critical
Value

μ_0
Value

Critical
Value

Sample Statistic

$$H_0: =$$
$$H_A: \neq$$

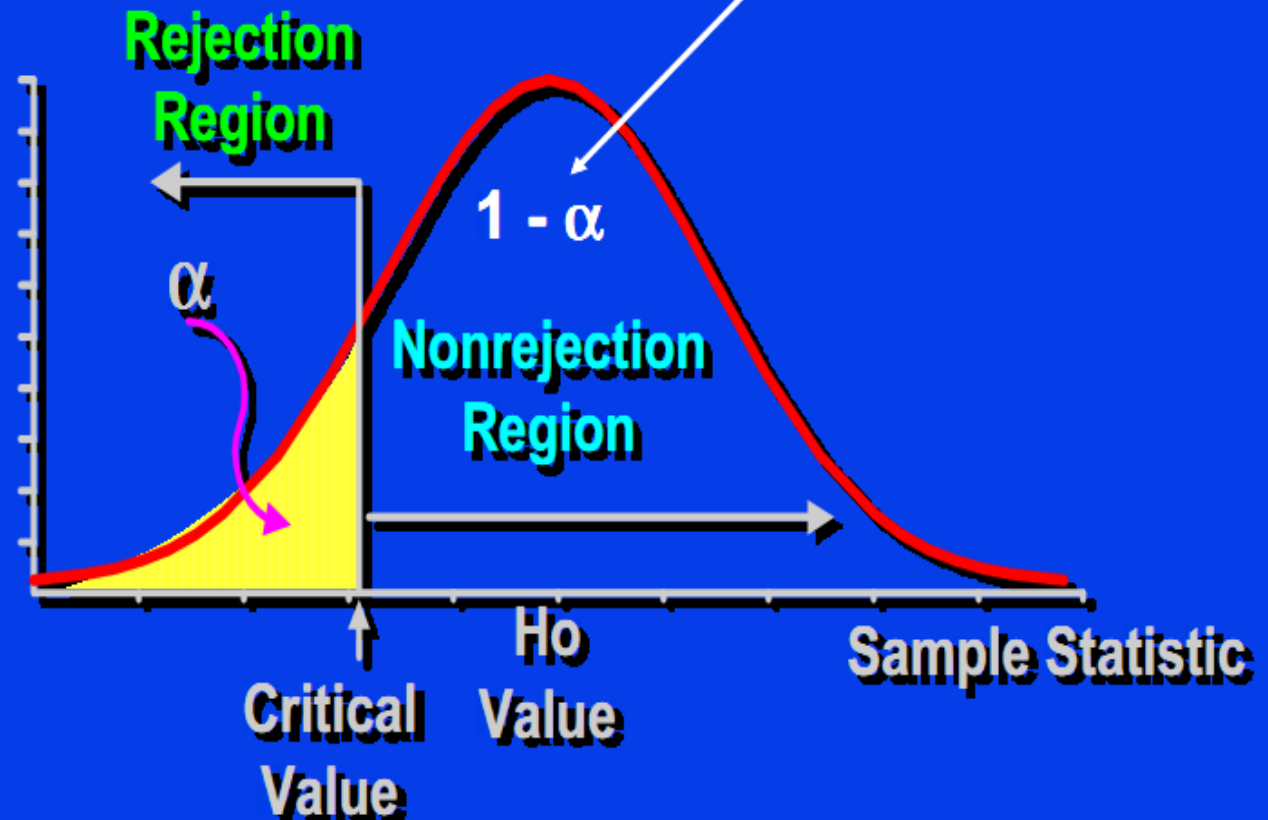
Rejection Region

(One - tail test: Left tail test)

Sampling Distribution

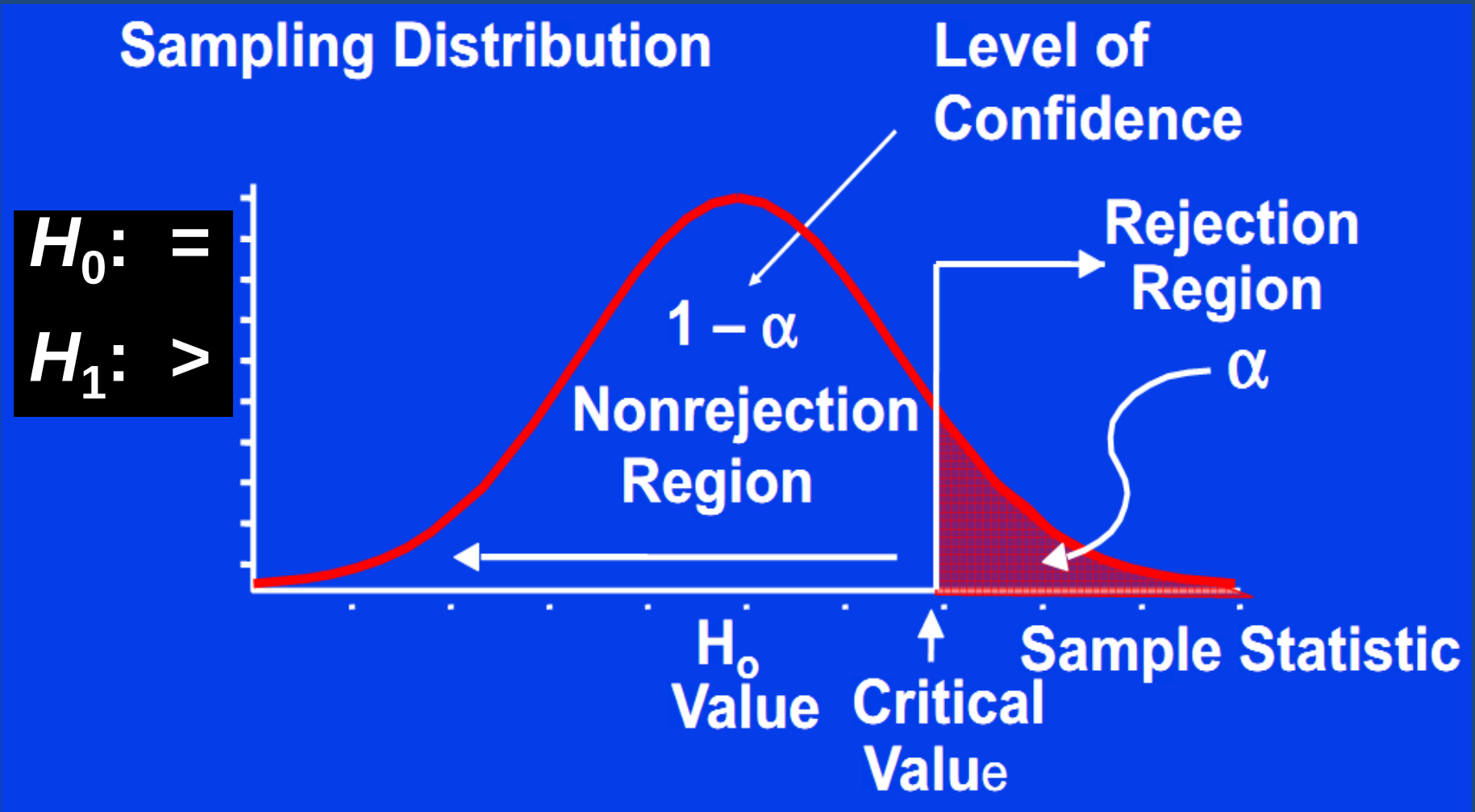
Level of Confidence

$$H_0: =$$
$$H_1: <$$



Rejection Region

(One - tail test: Right tail test)



3) Decide on the appropriate test statistic for the hypothesis (Z , t , χ^2 , F , etc) and compute the test statistic value

Cont.

Apply the formula of the test statistics to get the calculated value

$$\text{test statistic} = \frac{\text{Estimate-Hypothesized Parameter}}{\text{Standard Error}}$$

- Compare the calculated value to the tabulated or the critical value
- Check which zone the calculated value falls into

4) Make decision as per the decision rule

There are two possible decisions:

I) **Reject the null hypothesis**

- Conclude that there is enough evidence to support the alternative hypothesis

II) **Do not reject the null hypothesis**

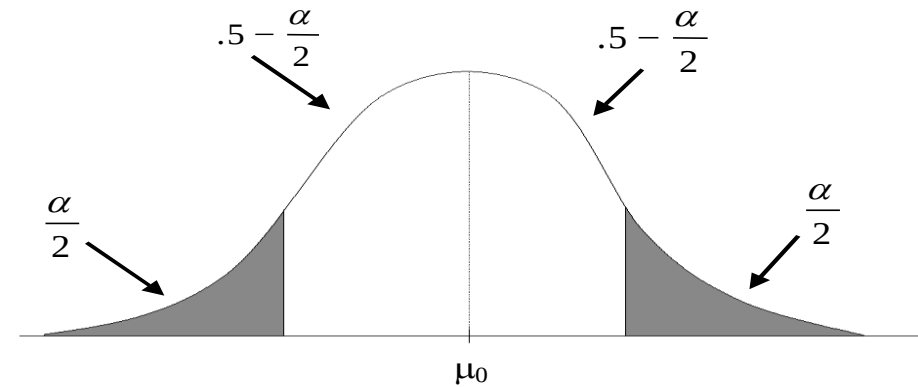
- Conclude that there is *not* enough evidence to support the alternative hypothesis
- If the numerical value of the test statistic **falls in the rejection region**, we reject the null hypothesis
- If the test statistic **does not fall in the rejection region**, we do not reject H_0

Decision rule

Form of H_a : $\mu \neq \mu_0$
2-tail hypothesis

If $|z| > |z_{\alpha/2}|$

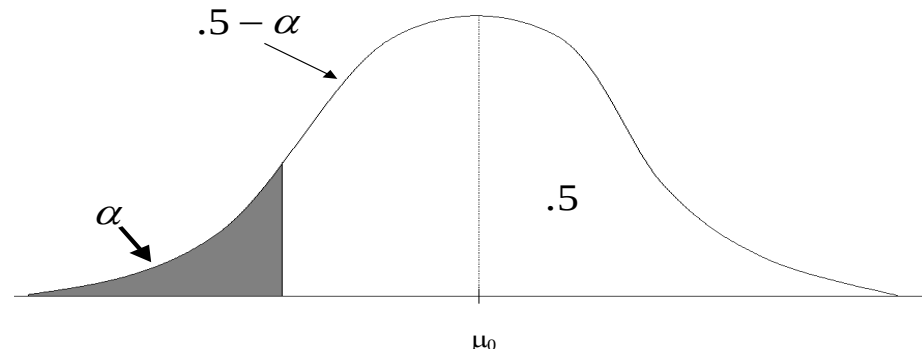
Then reject the null hypothesis.



Form of H_a : $\mu < \mu_0$
1-tail hypothesis

If $z < -z_{\alpha}$

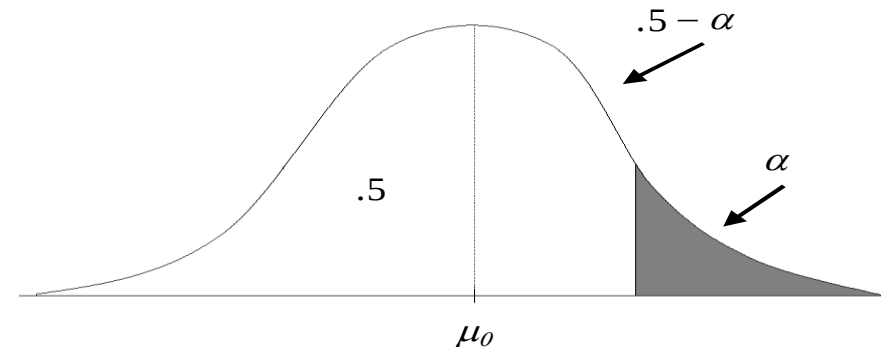
Then reject the null hypothesis.



Form of H_a : $\mu > \mu_0$
1-tail hypothesis

If $z > z_{\alpha}$

Then reject the null hypothesis



p-value

- **p-value:** is the probability of obtaining a test statistic at least as extreme ($\leq$ or $\geq$) as the observed sample value, given H_0 is true
 - Also called **observed level of significance**
 - Smallest value of α for which H_0 can be rejected
- **Decision :** Compare the p-value with α
 - If p-value $\geq \alpha$, do not reject H_0
 - If p-value $< \alpha$, reject H_0

Con.

- 5) draw conclusion

Errors in making Decision

1. Type I Error

- Probability of rejecting true null hypothesis
- Probability of accepting a false alternative hypothesis
- Probability of Type I Error is α (Alpha)
 - Called level of significance

2. Type II Error

- Probability of failing to reject a false null hypothesis
- Probability of rejecting a true alternative hypothesis
- Probability of Type II Error is β (Beta)

Decision Results

H_0 : Innocent

Jury Trial		
Verdict	Actual Situation	
	Innocent	Guilty
Innocent	Correct	Error
Guilty	Error	Correct

H_0 Test		
Decision	Actual Situation	
	H_0 True	H_0 False
Accept H_0	$1 - \alpha$	Type II Error (β)
Reject H_0	Type I Error (α)	Power ($1 - \beta$)

Type I error (false positive)



Factors Affecting Type II Error

- Significance level
 - β Increases when α decreases
- Population standard deviation
 - β Increases when σ increases
- Sample size
 - β Increases when n decreases

β ↑
 α ↓

β ↑
 σ ↑

β ↑
 n ↓

Power of a statistical test

- The power of a statistical test is the probability of rejecting H_0 , when H_0 is really false. Thus power = $1 - \beta$.
- Clearly if the test maximizes power, it minimizes the probability of Type 2 error β .

Hypothesis testing on a single population mean

EXAMPLE:

A researcher claims that the mean of the IQ for 64 students is 110 and the expected value for all population is 100 with standard deviation of 10.

Test the hypothesis that the population mean is different from 100 .

Solution:

1. $H_0: \mu = 100$ VS $H_A: \mu \neq 100$
2. Assume $\alpha = 0.05$, z-critical at 0,025 is equal to 1.96.
3. $z = (110 - 100) \cdot 8 / 10 = 8$
4. $8 \geq 1.96$ Therefore, the decision is reject the null hypothesis
5. We have sufficient evidence to support that the mean IQ of students is different from 100.

Example :

Researchers are interested in the mean level of some enzyme in a certain population. They are asking: can we conclude that the mean enzyme level in this population is different from 25? They collect a sample of size 10 from a normally distributed population with a known variance, $\sigma^2 = 45$. The calculated sample mean is = 22.

Step 1: $H_0: \mu = 25$

$H_A: \mu \neq 25$

Step 2: $\alpha = 0.05$

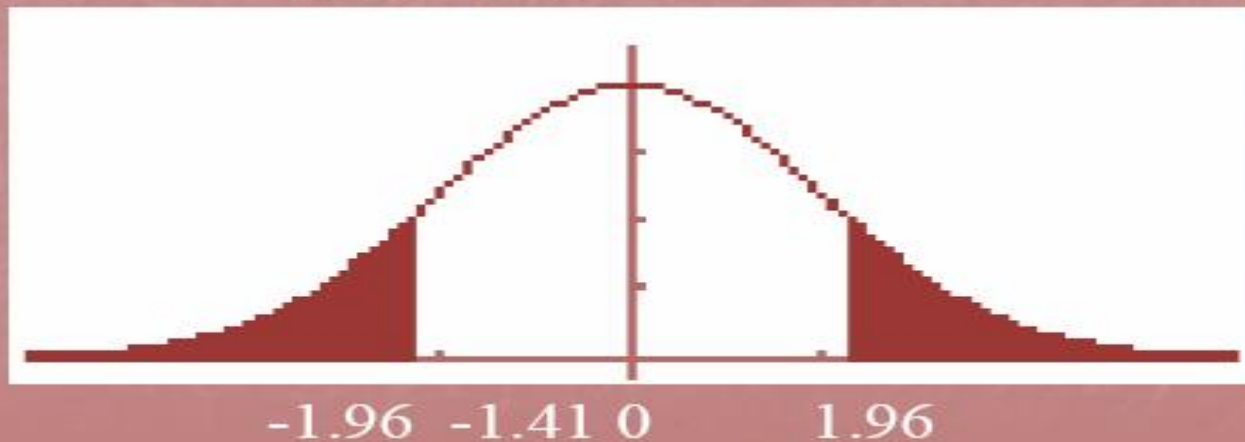
Step 3: Testing a hypothesis about population mean.

The population is normally distributed. Population variance is known

$$Z = (\bar{X} - \mu_0) / (\sigma / \sqrt{n})$$

■ Step 6: $\bar{X} = 22$, $\mu_0 = 25$, $\sigma = \sqrt{45}$, $n = 10$

$$\Rightarrow Z = \frac{22 - 25}{\sqrt{45 / 10}} = -1.41$$



Step 5: Since -1.41 falls in the acceptance region we accept the null hypothesis.

➤ The mean enzyme level in the population is not different from 25.

When small sample size and population variance is not given
the test statistic is:

$$t = \frac{\bar{X} - \mu_0}{S / \sqrt{n}}$$

Example: Serum amylase determination were made on a sample of 15 apparently health subjects. The sample yielded a mean of 96 units/100ml and a standard deviation of 35 units/100ml. The population was normally distributed but the variance was unknown. We want to know whether we can conclude that the mean of the population is different from 120.

Solution:

$$H_0: \mu = 120$$

$$H_A: \mu \neq 120$$

$$\diamond \text{ Mean} = 96 \quad \text{SD} = 35 \quad n = 15$$

$\diamond$ T-test is the appropriate test, since

✓ The sample size is small

✓ The population is normally distributed

✓ The population variance is unknown

$$t = \frac{\bar{x} - \mu_0}{S / \sqrt{n}}$$

- $\alpha = 0.05$

$$t_{0.025,14} = 2.1448$$

$$t = \frac{96 - 120}{35/\sqrt{15}} = -2.65$$

- Since $-2.65 < -2.1448$, it falls in the rejection region we reject the null hypothesis. The mean of the population from which the sample came is not 120.



Hypothesis testing for single proportions

Example: In the study of childhood abuse in psychiatry patients, Brown found that 166 in a sample of 947 patients reported histories of physical or sexual abuse.

- constructs 95% confidence interval
- test the hypothesis that the true population proportion is 30%

Solution (a)

- The 95% CI for P is given by:

$$\begin{aligned}\hat{p} \pm z_{\frac{\alpha}{2}} \sqrt{\frac{p(1-p)}{n}} \\ \Rightarrow 0.175 \pm 1.96 \times \sqrt{\frac{0.175 \times 0.825}{947}} \\ \Rightarrow 0.175 \pm 1.96 \times 0.0124 \\ \Rightarrow [0.151; 0.2]\end{aligned}$$

Solution (b)

To the hypothesis we need to follow the steps

Step 1: State the hypothesis

$$H_0: P = 0.3$$

$$H_a: P \neq 0.3$$

Step 2: Fix the level of significant ($\alpha=0.05$)

Step 3: Compute the calculated and tabulated value of the test statistic

$$Z_{cal} = \frac{\hat{p} - P_0}{\sqrt{\frac{p(1-p)}{n}}} = \frac{0.175 - 0.3}{\sqrt{\frac{0.175(0.825)}{947}}} = \frac{-0.125}{0.0124} = -10.08$$

$$Z_{tab} = 1.96$$

Step 4: Comparison of the calculated and tabulated values of the test statistic

- ❖ Since the tabulated value is smaller than the calculated value of the test then we reject the null hypothesis.

Step 5: Conclusion

- ❖ Hence we concluded that the proportion of childhood abuse in psychiatry patients is different from 0.3

Hypothesis testing for two sample means

- ❖ The steps to test the hypothesis for difference of means is the same with the single mean.

Step 1: state the hypothesis

$$H_0: \mu_1 - \mu_2 = 0$$

VS

$$H_A: \mu_1 - \mu_2 \neq 0, H_A: \mu_1 - \mu_2 < 0, H_A: \mu_1 - \mu_2 > 0$$

Step 2: Significance level (α)

Step 3: Test statistic

$$Z_{cal} = \frac{(\bar{x} - \bar{y}) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

Example

- ❖ A researchers wish to know if the data they have collected provide sufficient evidence to indicate a difference in mean serum uric acid levels between normal individual and individual with down's syndrome.
- ❖ The population data consists of serum uric acid readings on 12 individuals with down's syndrome and 15 normal individuals. The sample means are 4.5mg/100ml and 3.4 mg/100ml with population standard deviation of 2.9 and 3.5 mg/100ml respectively.

$$H_O : \mu_1 - \mu_2 = 0$$

$$H_A : \mu_1 - \mu_2 \neq 0$$

Solution

❖ The test statistic:

$$\begin{aligned} Z_{cal} &= \frac{(\bar{x} - \bar{y}) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} = \frac{(4.3 - 3.4) - 0}{\sqrt{\frac{2.9^2}{12} + \frac{3.5^2}{15}}} \\ &= \frac{1.6}{\sqrt{1.5178}} = \frac{1.6}{1.23} = 5.33 \end{aligned}$$

$$Z_{\frac{\alpha}{2}} = Z_{0.025} = 1.96$$

❖ Since $5.33 > 1.96$ reject the null hypothesis, i.e., there is an indication that the means are not equal.

Small sample size and population variance is not given

- The test statistic will be student's t-statistic with degree of freedom equals to $n_1 + n_2 - 2$
- Hence the tabulated value of t is read from the table.
- The decision remains the same

The test statistic is:

$$t_{\text{cal}} = \frac{\bar{X}_1 - \bar{X}_2}{S_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

$$S_p = \sqrt{\frac{(n_1 - 1)S_1^2 + (n_2 - 1)S_2^2}{n_1 + n_2 - 2}}$$

Example:

Serum amylase determinations were made on a sample of 15 apparently healthy subjects and 22 hospitalized subjects.

Suppose the samples yielded

- ❖ $\bar{x}_1 = 96$ units/100 ml $\bar{x}_2 = 120$ units/100 ml
- ❖ $s_1 = 35$ units/100 ml $s_2 = 40$ units/100 ml
- ❖ The populations are approximately normally distributed.
- ❖ Would we be justified in concluding that the population means are different?

Solution

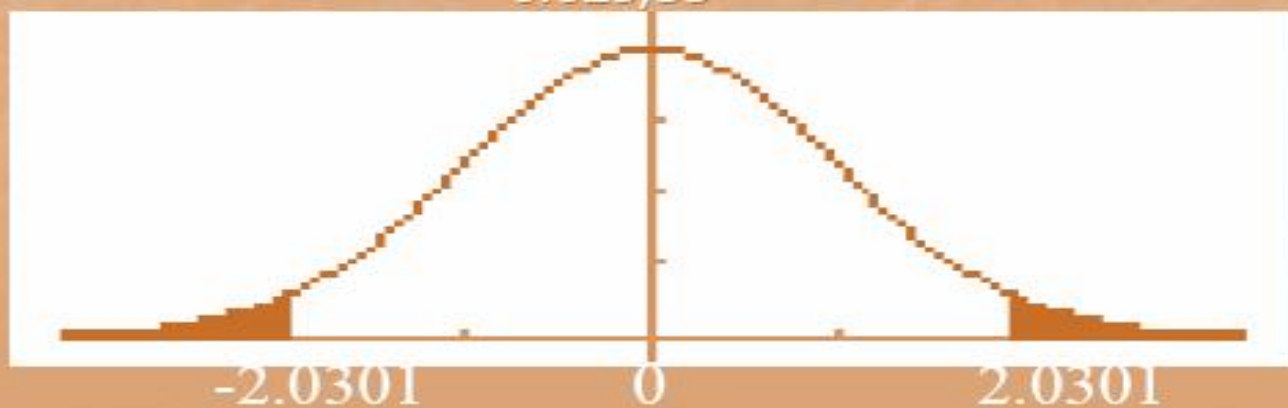
- $H_0: \mu_1 = \mu_2$

- $H_1: \mu_1 \neq \mu_2$

- t-test is the appropriate one,
with d.f. = $n_1 + n_2 - 2$.

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{S_P^2}{n_1} + \frac{S_P^2}{n_2}}}$$

- $\alpha = 0.05 \Rightarrow t_{0.025, 35} = 2.0301$



$$S_P = \sqrt{\frac{14 \times 35^2 + 21 \times 40^2}{15 + 22 - 2}} = 38.1$$
$$t = \frac{(96 - 120) - 0}{\sqrt{\frac{(38.1)^2}{15} + \frac{(38.1)^2}{22}}} = -1.88$$

- ❖ Since -1.88 falls in the acceptance region, we accept the null hypothesis.
- ❖ On the basis of these data we can not conclude that, the two population means are different.

Hypothesis testing for two population proportion

- ❖ Let n_1 and n_2 be the sample size from the two population.
- ❖ If x and y are the out come of interest then the point estimate for each population is given by $p_1=x/n_1$ and $p_2=y/n_2$ respectively.
- ❖ The point estimates of $\pi_1-\pi_2 = p_1-p_2$
- ❖ If the sample size is large and $n_1p_1>5$, $n_1(1-p_1)>5$, $n_2p_2>5$, $n_2(1-p_2)>5$
- ❖ The interval estimate for the difference of proportions is given by:

$$\left(p_1 - p_2 \pm z_{\frac{\alpha}{2}} \sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}} \right)$$

Hypothesis testing for two proportions

- ❖ To test the hypothesis

$$H_0: \pi_1 - \pi_2 = 0$$

VS

$$H_A: \pi_1 - \pi_2 \neq 0$$

- ❖ We use a pooled sample estimate for the common hypothesized proportion, which is a weighted average of the sample proportions, with the sample size as weights.

$$p = \frac{n_1 P_1 + n_2 P_2}{n_1 + n_2}$$

- The standard error of the difference is given by:

$$SE_{(P_1-P_2)} = \sqrt{\frac{P(1-P)}{n_1} + \frac{P(1-P)}{n_2}} = \sqrt{P(1-P)\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$$

- The test statistic becomes $Z =$

$$\frac{(P_1 - P_2) - (\pi_1 - \pi_2)}{SE_{(P_1-P_2)}}$$

Example

- ❖ Two hundred patients suffering from a certain disease were randomly divided into two equal groups.
- ❖ Of the first group, 78 recovered within three days. Out of the other 100, who were treated by a new method, 90 recovered within three days.
- ❖ The physician wished to know whether the data provide sufficient evidence to indicate that the new treatment is more effective than the standard.

Solution

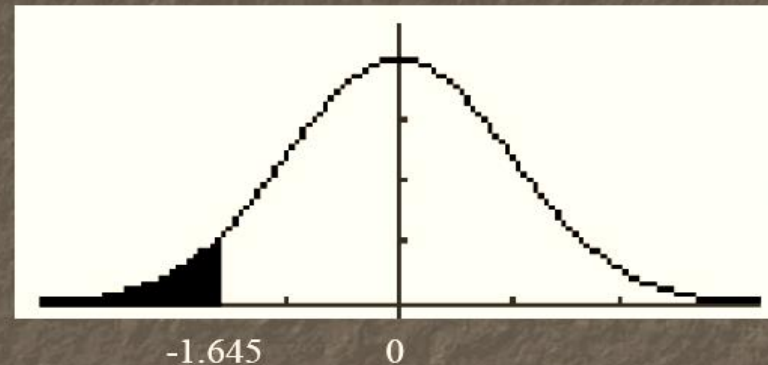
❖ $H_0 : \pi_1 = \pi_2$

❖ $H_1: \pi_1 < \pi_2$

❖ $P_1 = 78/100 = 0.78, P_2 = 90/100 = 0.90$

$$P = \frac{100(0.78) + 100(0.90)}{100 + 100} = 0.84$$

• $\alpha = 0.05 \Rightarrow Z_{0.05} = 1.645$



$$Z = \frac{0.78 - 0.90}{\sqrt{0.84(1-0.84)\left(\frac{1}{100} + \frac{1}{100}\right)}} = -2.32$$

- ❖ Since $-2.32 < -1.645$ it falls in the rejection region, we reject the null hypothesis.
- ❖ The data suggests that the new treatment is more effective than the standard.

□ Small sample size:

- If the sample size is small then use student's t-statistic at $n_1 + n_2 - 2$ degrees of freedom with the given level of significance.

Thank you!!!